Quiz 1

Q1. There are two types of data independence in database management systems: Logical Data Independence, and Physical Data Independence. Read the scenario below, then answer the following 3 questions.

You work for a music streaming company. The company has decided to add a new column in its database called Location where it will track each user's country. This information is not shown in the front end.

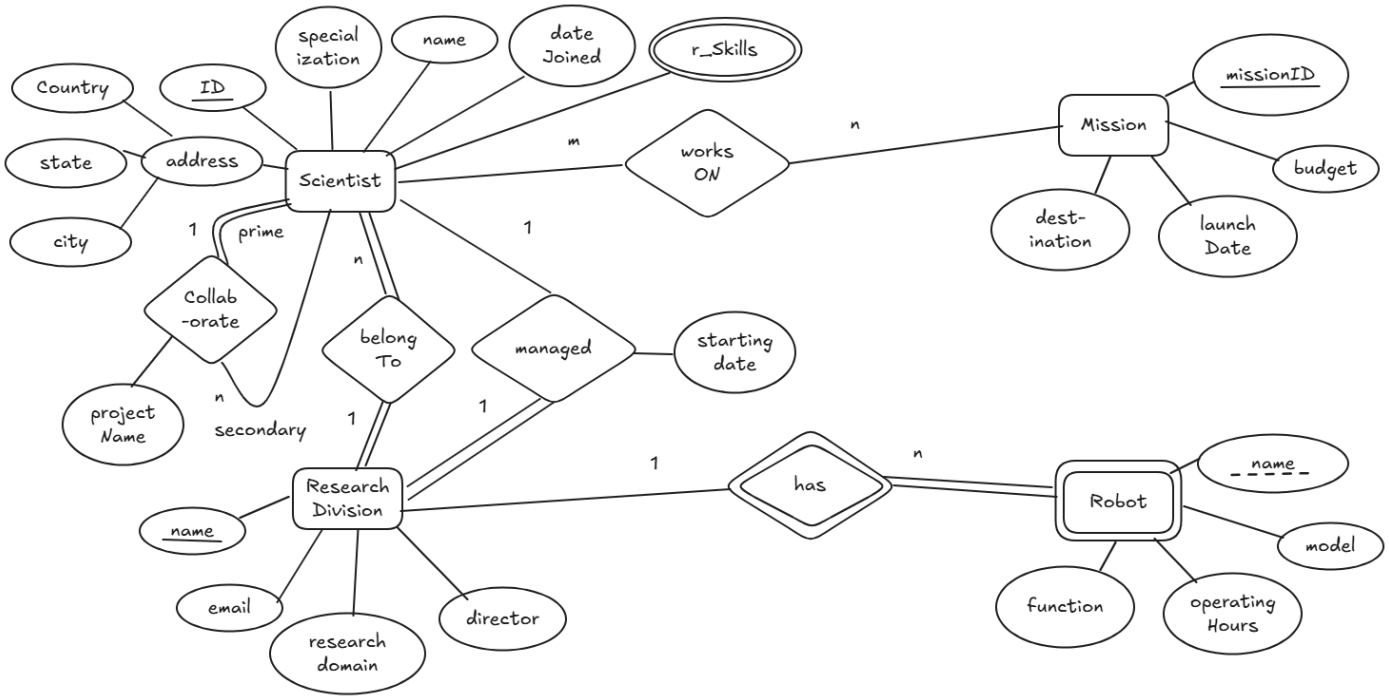
a) What type of Data Independence is this? **(1 Mark)**
b) Which Schema does this change in relation to Three Schema Architecture? **(1 Mark)**
c) Name the next step of the affected schema you mentioned in (b) and mention whether it will show any changes due to the given scenario. **(1 Mark)**

a) Ans: LDI
b) Ans: Conceptual Schema
c) Ans: External View, no changes

Q2. A Space Research Organization wants to create a database system for managing its missions, scientists, and facilities. The requirements are given below. **(7 Marks)**

- Scientists are part of a Research Division. Scientists have unique id, specialization, name, address, date_joined, and a list of research_skills.
- A Scientist's address is composed of country, state, and city.
- Each Research Division has a unique name, email, research_domain, and director.
- Each Scientist must belong to a Research Division, and each Research Division must have many scientists.
- Each Research Division is managed by one Scientist. A Scientist cannot manage more than one division. When a Scientist starts managing a Division, the starting_date is recorded.
- Scientists can collaborate with other Scientists. Each Scientist can collaborate with more than one colleague. A project name is recorded for each collaboration.
- Scientists also work on Space Missions. Each Mission has a unique mission_id, destination, budget, and launch_date.
- Some Research Divisions have Robots for assisting research. Each Robot has a name, model, function, and operating_hours. None of these attributes are guaranteed to be unique, different divisions may even have robots with the same name. However, note that the same Division will not reuse the same name for its own robots.

Now, draw an ER diagram of this system. Write down all assumptions (if any).


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city
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